

The UK Linguistics Olympiad 2022 - Round 2

Problem 1. Mapudungun (15 marks)

Mapudungun is an Araucanian language spoken by approximately 258,000 people (as of 2012) in Chile and Argentina.

Note that in Mapudungun, **y** is a vowel; **c, č, ɽ, θ, ɲ, ɳ** and **ɭ** are consonants; an acute accent as in **á** marks a stressed syllable.

Nouns in Mapudungun can be classified into two types: simplex nouns and nominal compounds. Here are some simplex nouns, with the stress marked and meanings given:

koɽkóma <i>coot</i>	anúɲka <i>plant</i>
ɲamýn <i>foot</i>	wanjilén <i>star</i>
mawíθa <i>woodland</i>	piýýɽka <i>flute</i>
elkoɲiwe <i>womb</i>	acuɽpén <i>floating ash</i>



1.1) On your answer sheet, assign stress to the following simplex nouns:

(a) yna <i>itch</i>	(c) ɽafken <i>sea</i>
(b) puɲpuja <i>armpit</i>	(d) piwke <i>heart</i>

Here are some compound nouns, with the stress marked, and both the meaning and literal translations given:

cafo-kučán <i>cold</i> (lit. 'cough-disease')	kuθi-fóro <i>spine</i> (lit. 'mortar-bone')
caɲúl-namýɲ <i>toe</i> (lit. 'finger-foot')	iló-čewa <i>dogmeat</i> (lit. 'meat-dog')
loɲkó-kacíɽa <i>head of wheat</i> (lit. 'head-wheat')	ɲé-čewa <i>dog's eye</i> (lit. 'eye-dog')
myta-píru <i>snail</i> (lit. 'horn-worm')	wecóθ-yu <i>nostril</i> (lit. 'hole-nose')

1.2) Assign stress to the following nominal compounds.

(a) mapu-ce <i>Mapuche people</i> (lit. 'land-people')	(e) wenu-mapu <i>heaven</i> (lit. 'high-land')
(b) ɽpi-acawaɽ <i>chicken wing</i> (lit. 'wing-chicken')	(f) foro-caɽwa <i>fishbone</i> (lit. 'bone-fish')
(c) kyčal-ko <i>petroleum</i> (lit. 'fire-water')	(g) loɲko-moyo <i>nipple</i> (lit. 'head-breast')
(d) θomo-lamɲen <i>sister</i> (lit. 'woman-sister')	(h) ciɽkatuwe-ruka <i>school</i> (lit. 'writing-house')

Below are some Mapudungun verb forms and their translations, with stress marked.

yčýfkelafíjmi <i>you do not usually throw it</i>	ɽánlawájmi <i>you will not die</i>
čipákelájmi <i>you do not usually go out</i>	yčýffín <i>I throw it</i>
θewmálawafíj <i>s/he will not make it</i>	amulájmi <i>you do not go</i>
ɽanýmkefín <i>I usually kill it</i>	yčýɲélajmi <i>you are not thrown</i>
leɲékelájmi <i>you are not usually watched</i>	lelifíjmi <i>you watch it</i>
amuýmfi <i>s/he causes it to come</i>	jékeláj <i>s/he does not usually bring</i>
θewmafín <i>I make it</i>	θewmaýmwfíj <i>s/he will cause it to make</i>

1.3) Assign stress to and translate the following verbs:

(a) leɲelawan	(d) jeɲekelawaj
(b) jeɲelajmi	(e) θewmakejmi
(c) čipalawan	(f) yčýfymkelawafíj

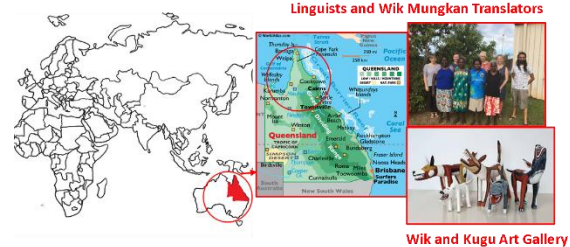
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Problem 2. Wik-Mungkan (15 marks)

Wik-Mungkan (literally: "to swallow one's words") is a Paman language spoken in Queensland, Australia by around 1,650 Wik-Mungkan people. During the dry season, fresh water is often in short supply and is considered valuable.



1. kek kuchek
2. kuchek thayan
3. ma' ek
4. ma' puk pi'an
5. ma' puuy
6. ma' thayan
7. min
8. ngak
9. ngak min
10. ngak way
11. ngangk
12. ngangk ek
13. ngangk min
14. ngangk thayan
15. ngangk way
16. puuy
17. puuy ek
18. thayan
19. wik thayan

- A. alcohol
- B. brave
- C. crab
- D. crab shell
- E. fingernail
- F. fresh water
- G. good
- H. handcuffs
- I. happy
- J. heart
- K. law
- L. sad
- M. shoulder blade
- N. strong, firm
- O. stubborn
- P. thumb
- Q. top of spear
- R. trustworthy (with things)
- S. water

2.1. Determine the correct correspondences.

2.2. Translate into English: (a) **kek**, (b) **puuy ngangk**, (c) **way**.

2.3. Translate into Wik-Mungkan: (a) *shell*, (b) *hand*, (c) *worried*.

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Problem 3. Niuean (20 marks)

Niuean is a Polynesian language spoken by nearly 8,000 people around the world. It is the official language of Niue, although most of its speakers live in other countries, such as New Zealand.

Below are some sentences in Niuean with a possible translation into English. Note that **ā** and **ū** are long vowels, and that **g** represents the *ng* sound in “sing”.



Niuean	English
Kua kai noa a au.	<i>I have only eaten.</i>
Kua fai fakatino foki ne tā e ia.	<i>There have also been pictures that he drew.</i>
Muhu moa tūmau.	<i>There are always plenty of birds.</i>
Ne fai faiaoga e kāmuta.	<i>The carpenter had teachers.</i>
Kua kitia e ia a au.	<i>He has seen me.</i>
To kai he moa ka holoholo e au e ika.	<i>The bird that I will wash will eat the fish.</i>
Ne totou a Sione.	<i>Sione read.</i>
Tā tūmau e Mele e fakatino.	<i>Mele is always drawing the picture.</i>
Ne kai e ika ne takafaga he tama	<i>The fish that the child caught ate</i>
To holoholo foki he tama e vaka ne tā he kāmuta.	<i>The child will also wash the canoe that the carpenter built.</i>
To muhu ika a Mele.	<i>Mele will have plenty of fish.</i>
Muhu tama foki e faiaoga ka kitia he moa.	<i>The teacher also has plenty of children that the bird will see.</i>
Fai vaka a Sione ne holoholo e au.	<i>Sione has canoes that I washed.</i>

3.1) On your answer sheet, translate the following sentences into English. If there are multiple possible translations, give only one.

- Fai moa noa.**
- Kua holoholo foki he faiaoga ne takafaga e au a ia.**
- To muhu vaka e tama ne kitia he moa ka holoholo e Sione.**

3.2) Translate the following sentences into Niuean.

- He will also read.*
- Sione has only had fish that the teacher will eat.*
- The teacher that Mele saw built the canoe.*
- There have always been plenty of carpenters.*

3.3) Explain your solution.

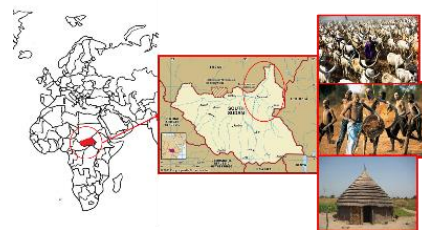
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Problem 4. Dinka (25 marks)

Dinka is a Nilotic dialect cluster with about 1.3 million native speakers, mostly ethnic Dinka people in South Sudan. There are several main varieties, but this problem focuses on the Agar dialect.



When linguists first studied the language, they believed that the singular and plural forms of Dinka nouns were completely unpredictable. More recently, however, studying the way that verbs conjugate in Dinka allowed linguists to find patterns in the singular and plural forms of nouns. This has allowed many nouns to be grouped according to declension patterns, although many remain unexplained.

Below are 22 Dinka nouns, in both singular and plural forms, each of which follow a one of the common patterns. The translations have been provided only for interest: they have no bearing on the solution to the problem.

Note: **t**, **d**, **n**, **j** and **ŋ** are consonants; **ɛ** is the vowel in *bed*, and **ɔ** is the vowel in *not*. Dinka has three vowel lengths: **a** short, **aa** medium, and **aaa** long; as well as three tones, **á** high, **à** low, and **â** falling.

Singular	Plural	Translation	Singular	Plural	Translation	Singular	Plural	Translation
láj	làaj	<i>animal</i>	màac	mêec	<i>fire</i>	ḑél	ḑèel	<i>path</i>
gâaar	gêer	<i>ankle bell</i>	rúp	rwòop	<i>forest</i>	wáal	wál	<i>plant</i>
ḡaaar	ḡóor	<i>bean</i>	àdjéel	àdíil	<i>gazelle</i>	ḡêeel	ḡéel	<i>python</i>
dít	djèet	<i>bird</i>	bôook	bóok	<i>hide</i>	dèenḡ	dêenḡ	<i>rain</i>
àḡwòòḡḡ	àḡóoḡḡ	<i>blacksmith</i>	ròok	ròok	<i>kidney</i>	àmàaal	àméel	<i>sheep</i>
ḡàanḡ	ḡêenḡ	<i>bow, gun</i>	ḡòòk	ḡòòk	<i>louse</i>	àtwòoor	àtúur	<i>slime</i>
gól	gàal	<i>cowdung fire</i>	àḡâaaḡḡ	àḡééḡḡ	<i>monitor lizard</i>			
twóòḡḡ	tónḡḡ	<i>egg</i>	àḡòòòk	àḡóòk	<i>monkey</i>			

Here are some forms of 4 regular Dinka verbs.

Root	1 st person	3 rd person	Translation
nòḡḡ	nàanḡḡ	nòòḡḡ	<i>to have</i>
kùc	kwòoc	kùuc	<i>to not know</i>
màat	màaat	mèèet	<i>to smoke</i>
lòòk	làaak	lòòk	<i>to wash</i>

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4.1) Some singular or plural forms of Dinka nouns are given below. Only two of them follow one of the common patterns demonstrated above. On your answer sheet, mark which two they are.

Singular	Plural	Translation
àdèen		<i>beautiful one</i>
miiit		<i>firefly</i>
wèeet		<i>metal</i>
	tôon	<i>pot</i>
	jâak	<i>evil spirit</i>
	tûun	<i>horn</i>

4.2) Assuming that the following verbs conform to the previous pattern, fill in the correct form on your answer sheet:

Root	1 st person	3 rd person	Translation
a)	lwòccj	b)	<i>to be different</i>
c)	d)	cèem	<i>to eat</i>
e)	pèeεc	f)	<i>to loot</i>
wic	g)	wiic	<i>to need</i>
h)	i)	bòok	<i>to throw at</i>

4.3) Below are the singular or plural forms of 10 more Dinka nouns. By using the common patterns demonstrated before, predict the missing forms on your answer sheet. If there is more than one possible prediction, give them all.

Singular	Plural	Translation	Singular	Plural	Translation
ràaan	(j)	<i>person</i>	kók	o)	<i>hole in tree</i>
k)	wiil	<i>bristle</i>	p)	léek	<i>pestle</i>
rèeεc	l)	<i>fish</i>	ról	q)	<i>voice</i>
m)	kàal	<i>hole in ground</i>	jiiṭ	r)	<i>well</i>
n)	rím	<i>blood</i>	ànâaar	s)	<i>buffalo</i>

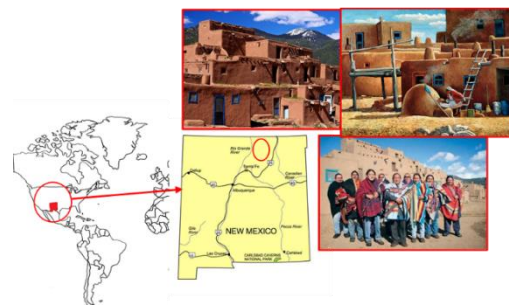
4.4) Explain your solution.

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Problem 5. Taos (25 marks)

Taos is a Tanoan language spoken by about 800 people in New Mexico.

Below are some sentences in Taos and their English translations. Note that **a**, **e**, **i**, **o** and **u** are nasal vowels, while **p**, **t** and **y** are consonants. Taos is a tonal language, but here tones have been omitted for simplicity. Note also that here “deer” is used exclusively to refer to one deer, while “deers” is used to refer to multiple deers.



Taos	English
kimpɛnemə ipihumɨ	<i>Our deers saw the cedar.</i>
kəkana pɛxi	<i>Your mother waited for the deer.</i>
imhɨnemə t'oymət'ɛ	<i>Their cedar hit the person.</i>
iwkine iwp'imət'ɛ	<i>Their blankets hit the heads.</i>
kip'oyna ip'jwxi	<i>Our brother waited for the sparrows.</i>
kəwcine ip'oymɨ	<i>Your eyes saw the brother.</i>
t'oyna ikixi	<i>The person waited for the blanket.</i>
kanemə ip'jwmət'ɛ	<i>The mothers hit the sparrow.</i>
əmcinemə ip'oymɨ	<i>His eye saw the brothers.</i>
əmp'oynemə iwcimɨ	<i>His brothers saw the eyes.</i>
kine ipipɛmət'ɛ	<i>The blankets hit the deers.</i>

5.1) Translate the following into English:

- imt'oynemə iwhumɨ**
- əpɛna kamət'ɛ**
- əwhɨne ipit'oymət'ɛ**
- p'jwnemə ipipɛxi**

5.2) Translate the following into Taos:

- Your sparrows waited for the mothers.*
- The heads saw the blanket.*
- Our eyes saw the heads.*
- Their deer waited for the brother.*

5.3) Explain your solution.

NB: some of the Taos sentences above would be considered odd by a native Taos speaker, who would use a passive construction instead. For simplicity, an active construction has been used here in all cases.